Decision Log

A running list of the important choices, the alternatives I considered, and why I picked what I did.

1. Score-based rules vs. a trained ML model

- **Chose:** a transparent additive risk score (0–100) from tunable weights.
- **Alternatives:** logistic regression / gradient boosting on the labelled data.
- **Why:** the brief demands that (a) a non-technical person can change thresholds without code, and (b) every decision has a plain-English reason. A hand-weighted score gives both for free. A trained model is my *next* step (see memo §5), but it would start as a black box and needs guardrails before it touches real orders.

2. Where the LLM is used — and where it is NOT

- **LLM used for:** judging address completeness/realness (incl. Hinglish, landmark-only addresses) and flagging fake names.
- **LLM NOT used for:** RTO probability (lookup + arithmetic), phone validity (regex), geography risk (data table), and the final money decision (expected-value maths).
- **Why:** LLMs are slow, cost money per call, and are non-deterministic. I only pay for one where language understanding genuinely beats code. This line is enforced in `engine/llm_check.py`.

3. Don't auto-cancel risky orders — the economics say so

- **Chose:** cancel only near-certain returns (score ≥ 85, ~0.2% of orders).
- **Alternative:** Head of Ops wanted to auto-cancel anything risky / stop COD in bad pincodes.
- **Why:** a delivered order earns ₹402; a return costs ₹169. Break-even RTO probability is **~70%**. Cancelling a 40–50% RTO order destroys more value (lost profit on the ones that would deliver) than it saves. The data confirmed it: holding below score 85 produced a *negative* net.

4. Thresholds calibrated against the real data, not guessed

- **Chose:** ship<20, confirm<45, nudge<85, else hold.
- **Why:** my first guess (35/55/75) put the hold bucket at 60% RTO, which *lost* ₹887 because it cancelled genuine orders. I swept thresholds against actual bucket RTO and breakeven economics and moved them so the hold bucket is ~100% RTO. This iteration is the whole point — the first guess was wrong and the data fixed it.

5. State-level geography, not pincode

- **Chose:** state risk table.
- **Alternative:** pincode-level risk.
- **Why:** only 1 pincode in the sample had ≥8 orders — pincode data is far too sparse to trust. State-level is statistically solid (Bihar 53% on 146 orders vs TN 20% on 158). The config still *allows* pincode overrides when real volume justifies it.

6. Marketing guardrail for paid-ads first orders

- **Chose:** never auto-cancel a first order from paid sources (meta/google/influencer) unless score ≥ 85.
- **Why:** directly answers Marketing's ₹300 CAC concern. A risky paid-ads newbie gets nudged to prepaid instead of cancelled.

7. Hinglish as the default WhatsApp language

- **Chose:** Hinglish messages by default (config-switchable to English/Hindi).
- **Why:** most Indian COD buyers read Hinglish fastest; higher reply rate = more RTOs caught.

8. Clean the dirty labels, don't drop silently

- The file had `Rto`, `delivered`, one `IN_TRANSIT`. I normalise case and keep only DELIVERED/RTO for scoring, and flagged this in the memo as a data-quality risk.

9. Conservative money assumptions, stated openly

- Confirmation prevents 50% of would-be RTOs; 50% of nudged customers accept prepaid.
- **Why:** better to under-promise savings. Both live as constants and are easy to update once real outcomes arrive.

10. Impact figures must carry an explicit time basis (correction)

- **What was wrong:** an earlier version reported the net impact as "~₹43,000/month" by scaling the 2,017-order sample up to 3,000 orders as if the sample were one month. But the sample spans ~6 months (1 Apr – 30 Sep 2026), so that conflated a six-month total with a one-month framing — the rupee numbers did not reconcile.
- **The fix:** `analysis/backtest.py` now derives `months_in_sample` from the data's own min/max dates and reports three reconciling figures: net over the sample (₹29,191), net per month (= net ÷ months ≈ ₹4,882), and net per 1,000 orders (≈ ₹14,472, which scales to any future volume). No single-month figure is ever implied from six months of data.
- **Lesson:** every impact number must state the window of its input data, the window of its claim, and the conversion between them. The backtest that produces these

numbers is now a saved, reproducible script in the repo rather than an inline calculation.